

RELATIONSHIP BETWEEN COUNTERMOVEMENT JUMP PERFORMANCE AND MULTIJOINT ISOMETRIC AND DYNAMIC TESTS OF STRENGTH

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ABSTRACT

The purpose of this investigation was to determine the relationship between countermovement vertical jump (CMJ) performance and various methods used to assess isometric and dynamic multijoint strength. Twelve NCAA Division I-AA male football and track and field athletes (age, 19.83 ± 1.40 years; height, 179.10 ± 4.56 cm; mass, 90.08 ± 14.81 kg; percentage of body fat, $11.85 \pm 5.47\%$) participated in 2 testing sessions. The first session involved 1 repetition maximum (1RM) (kg) testing in the squat and power clean. During the second session, peak force (N), relative peak force ($\text{N}\cdot\text{kg}^{-1}$), peak power (W), relative peak power ($\text{W}\cdot\text{kg}^{-1}$), peak velocity ($\text{m}\cdot\text{s}^{-1}$), and jump height (meters) in a CMJ, and peak force and rate of force development (RFD) ($\text{N}\cdot\text{s}^{-1}$) in a maximal isometric squat (ISO squat) and maximal isometric mid-thigh pull (ISO mid-thigh) were assessed. Significant correlations ($P \leq 0.05$) were found when comparing relative 1RMs (1RM/body mass), in both the squat and power clean, to relative CMJ peak power, CMJ peak velocity, and CMJ height. No significant correlations existed between the 4 measures of absolute strength, which did not account for body mass (squat 1RM, power clean 1RM, ISO squat peak force, and ISO mid-thigh peak force) when compared to CMJ peak velocity and CMJ height. In conclusion, multijoint dynamic tests of strength (squat 1RM and power clean 1RM), expressed relative to body mass, are most closely correlated with CMJ performance. These results suggest that increasing maximal strength relative to body mass can improve performance in explosive lower body movements. The squat and power clean, used in a concurrent strength and power training program, are recommended for optimizing lower body power.

KEY WORDS force, squat, power clean

INTRODUCTION

Previous investigations have compared performance in the countermovement vertical jump (CMJ) to strength and power in single-joint isometric (ISO) tests (1,3,13,20,25), multijoint ISO tests (11,12,16,22,28,30), and multijoint dynamic tests (5,7,27,29,30). Typically, CMJ performance is described by the following variables: peak force (N), relative peak force ($\text{N}\cdot\text{kg}^{-1}$), peak power (W), relative peak power ($\text{W}\cdot\text{kg}^{-1}$), peak velocity ($\text{m}\cdot\text{s}^{-1}$), and jump height (meters) (7,11,12,16,22). ISO testing variables are normally reported as peak force (N), relative peak force ($\text{N}\cdot\text{kg}^{-1}$) and rate of force development (RFD) ($\text{N}\cdot\text{s}^{-1}$), and dynamic strength tests are defined as a 1 repetition maximum (1RM) (kg) or a relative 1RM ($\text{kg}\cdot\text{kg}^{-1}$) (7,11,12,16,22,29,30). The unique aspect of this study is that no previous study that used the ISO and dynamic forms of the squat and power clean has compared all these variables in the context of a single investigation.

Investigations that have assessed strength (1,20) and RFD (3,24) in the ISO knee extension have not found significant correlations with CMJ height. The results from these investigations may suggest that single-joint isometric tests are poor predictors of performance in multijoint explosive movements such as a CMJ. More recent investigations have explored multijoint ISO tests and their relationships with CMJ performance (11,12,16,22,28,30). A CMJ requires the use of muscles that create movement across multiple joints; thus, it may be assumed that multijoint ISO tests can provide a better indication of the ability to create power in an explosive lower body movement such as a vertical jump. The ISO squat (28,30) and ISO mid-thigh pull (ISO mid-thigh) (11,12,16,22) are 2 methods that have been used to assess multijoint ISO strength and power. To our knowledge, only 2 investigations have assessed the relationship between the ISO squat and CMJ performance (28,30). These investigations have only provided a limited number of comparisons between the variables that define performance in an ISO squat and a CMJ. The results from these investigations have been conflicting as significant and nonsignificant correlations between the 2 tests have been reported (28,30). The ISO variation of the power clean, the ISO mid-thigh, has been compared to CMJ performance in

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TABLE 1. Performance variables for strength and power measurements in the isometric (ISO) squat, ISO mid-thigh pull, squat 1RM, power clean 1RM, and CMJ.

Isometric squat		ISO mid-thigh pull	
ISO RFD ($\text{N}\cdot\text{s}^{-1}$)	6102.0 ± 1579.0	ISO RFD ($\text{N}\cdot\text{s}^{-1}$)	3555.8 ± 1025.7
ISO peak force (N)	3522.2 ± 635.4	ISO peak force (N)	3143.8 ± 792.0
ISO peak force/BM ($\text{N}\cdot\text{kg}^{-1}$)	39.6 ± 4.6	ISO peak force/BM ($\text{N}\cdot\text{kg}^{-1}$)	35.2 ± 6.8
Squat 1RM		Power clean 1RM	
1RM (kg)	170.8 ± 22.6	1RM (kg)	113.1 ± 13.5
1RM/BM ($\text{kg}\cdot\text{kg}^{-1}$)	1.93 ± 0.21	1RM/BM ($\text{kg}\cdot\text{kg}^{-1}$)	1.28 ± 0.11
CMJ			
Peak power (W)	6437.1 ± 1046.3		
Relative peak power ($\text{W}\cdot\text{kg}^{-1}$)	72.6 ± 9.0		
Peak force (N)	1991.3 ± 338.4		
Relative peak force ($\text{N}\cdot\text{kg}^{-1}$)	22.4 ± 2.4		
Peak velocity ($\text{m}\cdot\text{s}^{-1}$)	3.66 ± 0.26		
Jump height (m)	0.56 ± 0.08		

Values are expressed as mean $\pm$ SD ($n = 12$).

ISO = isometric; 1RM = 1 repetition maximum; CMJ = countermovement vertical jump; RFD = rate of force development; BM = body mass.

several investigations (11,12,16,22). These investigations have provided numerous comparisons between the variables that define performance in an ISO mid-thigh and CMJ. Some studies have reported significant correlations between the variables that define these 2 tests (11,16,22), while other studies have reported nonsignificant correlations between the same variables (11,12,16). Thus, it appears that more investigation is necessary to clarify the relationships between multijoint ISO tests and the CMJ.

The most consistent results are present when comparing squat 1RM to CMJ performance. Significant correlations have been reported when assessing the relationships between squat 1RM and CMJ height (5,27,29). Additionally, when comparing squat 1RM tests to CMJ performance, Carlock et al. (7) found strong correlations between the following variables: squat 1RM and CMJ peak power, relative squat 1RM and CMJ and relative peak power, and CMJ height. Only Young and Bilby (31) demonstrated a nonsignificant correlation when comparing squat 1RM to CMJ height. Similar to the squat 1RM, the relationships between Olympic weightlifting movements (i.e., snatch and clean and jerk) and CMJ performance have been reported as strong (7). However, there appears to be a paucity of research when considering standard power cleans from the floor and CMJ performance.

The relationships between strength assessments and performance are of importance to both researchers and strength and conditioning practitioners. Although these relationships are not the source for cause and effect, they do provide researchers with a better understanding of the use of appropriate tests when assessing the effectiveness of

experimental treatments on strength, power, and other performance measures. Additionally, a more thorough understanding of strength assessments and their relationships with lower body force, velocity, and power may provide additional evidence regarding the incorporation of certain training methodologies. More specifically, practitioners may gain further insight into the biomechanical principles behind strength and power training.

The values reported for correlations between ISO and dynamic measures of strength and CMJ performance are quite diverse, and the relationships between these measures are still unclear and require further investigation. Additionally, many of the relationships between performance variables in these tests have yet to be investigated. Therefore, the purpose of this study was to assess the relationships between CMJ performance and multijoint ISO and dynamic tests of strength. From these relationships, there was an additional purpose of this investigation to dissect the role of strength and power in training programs aimed at improving performance in explosive lower body movements.

METHODS

Experimental Approach to the Problem

In order to assess the relationships between CMJ performance (CMJ peak force, peak power, peak velocity, and height) and multijoint ISO tests (ISO squat and ISO mid-thigh peak force, relative peak force, and RFD) and multijoint dynamic tests (squat and power clean 1RM and relative 1RM) of strength, descriptive correlations were used. Subjects participated in 2 testing sessions. The first session involved

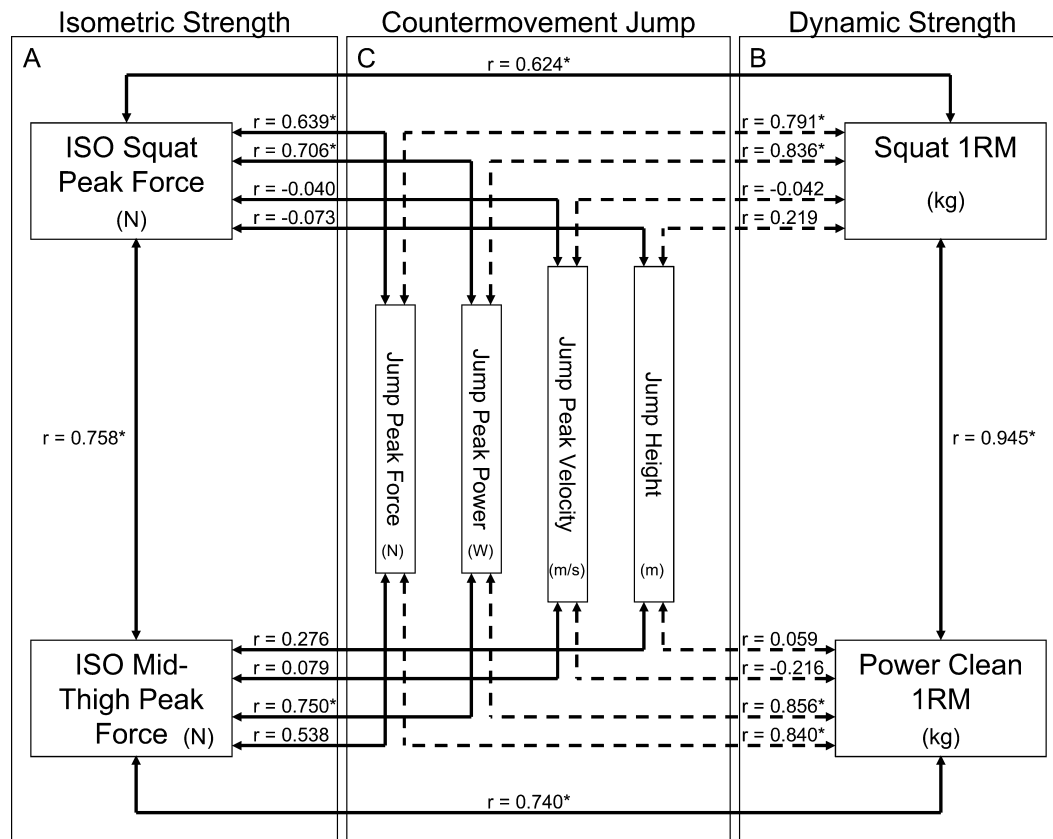


Figure 1. Correlations between absolute strength and power measurements in the isometric (ISO) squat, ISO mid-thigh pull, squat 1 repetition maximum (1RM), power clean 1RM, and countermovement vertical jump (CMJ). *Significant at $p \leq 0.05$.

squat 1RM and power clean 1RM testing. The second testing session assessed performance in the CMJ, ISO squat, and ISO mid-thigh. The testing sessions were separated by at least 4 days. The hypothesis was that multijoint ISO and dynamic tests of strength and relative strength would significantly correlate with CMJ performance variables.

Subjects

Twelve Division I-AA male athletes participated in this study (age, 19.83 ± 1.40 years; height, 179.10 ± 4.56 cm; mass, 90.08 ± 14.81 kg; % body fat, $11.85 \pm 5.47\%$; squat 1RM, 170.38 ± 21.72 kg; power clean 1RM, 112.50 ± 13.15 kg; squat 1RM to body mass ratio, 1.91 ± 0.22 ; power clean 1RM, 1.28 ± 0.11). The sample was composed of 7 football players and 6 track and field athletes (2 triple jumpers, 2 sprinters, and 2 throwers). These athletes were recruited because of the explosive nature of their sports and their past experience (≥ 4 years) with strength and power exercises. All subjects were tested during the off-season of their respective sports. Subjects abstained from vigorous training during the 3 days prior to the first testing session. Furthermore, subjects were asked to maintain their normal level of daily activity and their current

nutritional habits for 3 days prior to the first testing session and the 4–7 days that separated the first and second testing sessions. Approval from the Appalachian State University Institutional Review Board was obtained prior to the investigation. Study procedures were described to each subject, and each subject signed a consent form before participating in any testing.

Squat and Power Clean 1RM Testing

Subjects were tested for their squat 1RM and power clean 1RM during the initial testing session. A 5-minute cycle ergometer warm-up was performed prior to the 1RM testing protocol as described previously (26). The 1RM tests were completed in a randomized order, and a 30-minute rest period was provided between the 2 tests. Squats were performed so that the depth corresponded to a 70° knee angle. Knee angle was visually monitored by the same investigator for all testing sessions to ensure consistency in squat depth. If the appropriate depth was not achieved, the test was considered unsuccessful. Power clean attempts were considered successful when the catch of the bar was made followed by complete extension of the knees and hips so that

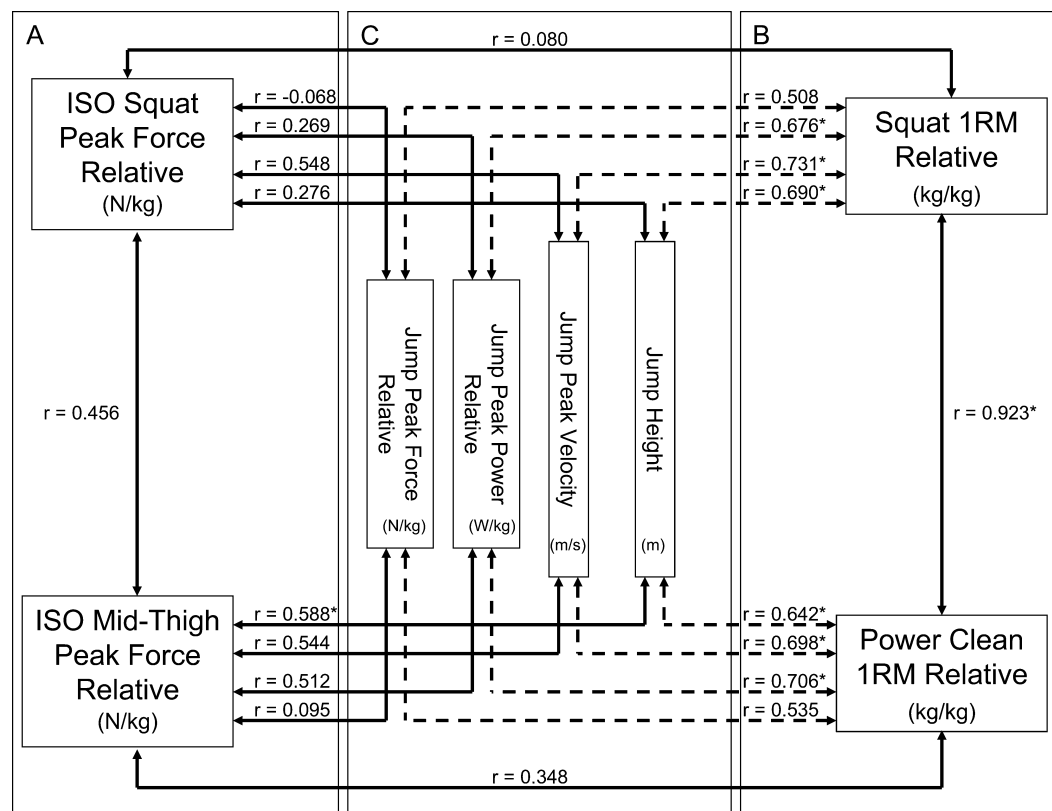


Figure 2. Correlations between relative strength and power measurements in the isometric (ISO) squat, ISO mid-thigh pull, squat 1 repetition maximum (1RM), power clean 1RM, and countermovement vertical jump (CMJ). *Significant at $p \leq 0.05$.

a fully erect position was attained. Strong verbal encouragement was given to all subjects during all 1RM attempts. Performance was analyzed as both a 1RM and a relative 1RM.

CMJ

After completing a 5-minute cycle ergometer warm-up, subjects performed CMJs on a force plate (AMTI, BP6001200, Watertown, MA). Subjects were permitted a minimum of 2 trials, and a 2-minute rest period was provided between trials. During the CMJ, peak force, peak power, peak velocity, and jump height were determined. Subjects were required to hold a plastic bar across their shoulders and were instructed to keep constant downward pressure on the bar throughout the jump so that the bar would not move independently of the body. The bar acted to counterweight the pull of 2 linear position transducers (LPTs) (PT5A-150; Celesco Transducer Products, Chatsworth, CA), resulting in a zero load. The 2 LPTs, located anterior and posterior to the subject, were attached to the bar. This resulted in the formation of a triangle that allowed the calculation of vertical and horizontal displacements through trigonometry involving constants and displacement

measurements (8). Signals from the 2 LPTs and the force plate underwent rectangular smoothing with a moving average half-width of 12. The analog signals were collected for every trial at 1000 Hz using a BNC-2010 interface box with an analog-to-digital card (NI PCI-6014; National Instruments, Austin, TX). Peak force, peak power, peak velocity, and jump height were all measured during the concentric phase of the CMJ. The start of the concentric phase was determined as the point at which the displacement-time curve became positive and was considered finished when the force-time curve became zero. Jump depth was self-selected by the subjects in order to maximize CMJ height. The maximum force recorded from the force-time curve during the concentric phase was reported as the peak force. Concentric peak velocity was measured as the change in bar displacement divided by the change in time. Concentric peak power was determined as the force multiplied by the velocity. Jump height was determined to be the difference between the maximum bar displacement and the bar displacement while in the standing position. Furthermore, peak force and peak power were analyzed relative to each subject's body mass. Specifically designed

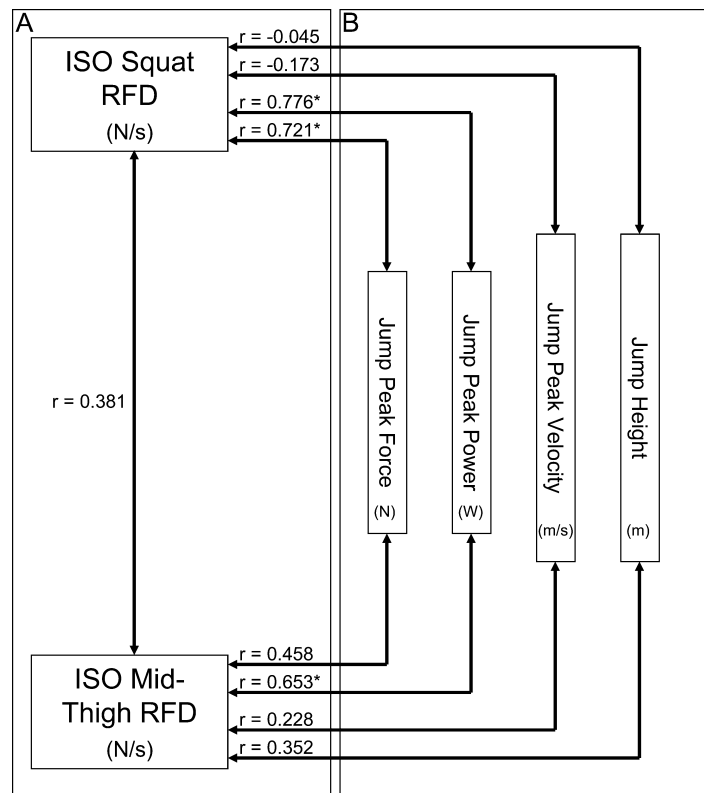


Figure 3. Correlations between rate of force development (RFD) in the isometric (ISO) squat and ISO mid-thigh pull and countermovement vertical jump (CMJ). *Significant at $p \leq 0.05$.

measured using a goniometer and standardized at 140° based on the reports of previous research (11,12,16,22). For this test, an immovable bar was positioned just below the crease of the hip. After the test administrator's verbal instruction, subjects pulled against the bar with maximal effort as quickly as possible (4). For the ISO squat, a knee angle of 140° was also used because it is closely associated with the production of maximal force when assessing ISO squats at various knee angles (21). Following the test administrator's verbal instruction, subjects pushed with maximal effort as quickly as possible against the immovable bar that was located on their shoulders. Each maximal ISO trial was performed for 3 seconds, and all subjects were given strong verbal encouragement during each trial. The maximum force recorded from the force-time curve during the 3-second ISO trial was reported as the peak force. This peak force was also used in the measurement of

LabVIEW (Version 7.1, National Instruments) programs were used for recording and analyzing the specific variables listed above.

ISO Testing

ISO testing was performed on a force plate located inside a power rack that used pins and hydraulic jacks to establish the desired testing position. For the ISO mid-thigh, knee angle was

relative peak force, which accounts for subject body mass. The RFD was determined as the peak force when a plateau in the force-time curve was achieved divided by the time in which it took to accomplish the given force level.

Statistical Analyses

Data were analyzed with descriptive statistics, and results are summarized as mean $\pm$ SD. Similar to previous investigations assessing CMJ performance and tests of strength, Pearson product-moment correlation coefficients were used to describe the relationships between variables (7,12,16,22, 28,29). The significance level for all statistics was set at $p \leq 0.05$. Typically, a statistical power ≥ 0.80 is considered acceptable. The correlations found to be statistically significant in this investigation had an average statistical power of 0.83 (range, 0.56–1.00). Kawomori et al. (16)

TABLE 2. Correlation between variables from countermovement jump (CMJ) in Figure 1C.

Variable	CMJ height (m)	CMJ PV ($\text{m}\cdot\text{s}^{-1}$)	CMJ PP (W)	CMJ PF (N)
CMJ height (m)	1.000*	0.826*	0.266	-0.116
CMJ PV ($\text{m}\cdot\text{s}^{-1}$)	0.826*	1.000*	0.004	-0.343
CMJ PP (W)	0.266	0.004	1.000*	0.878*
CMJ PF (N)	-0.116	-0.343	0.878*	1.000*

PV = peak velocity; PP = peak power; PF = peak force.
*Correlations significant at $p \leq 0.05$.

TABLE 3. Correlation between variables from countermovement jump (CMJ) in Figure 2C.

Variable	CMJ height (m)	CMJ PV ($\text{m}\cdot\text{s}^{-1}$)	Relative CMJ PP ($\text{W}\cdot\text{kg}^{-1}$)	Relative CMJ PF ($\text{N}\cdot\text{kg}^{-1}$)
CMJ height (m)	1.000*	0.826*	0.726*	0.291
CMJ PV ($\text{m}\cdot\text{s}^{-1}$)	0.826*	1.000*	0.693*	0.289
Relative CMJ PP ($\text{W}\cdot\text{kg}^{-1}$)	0.726*	0.693*	1.000*	0.779*
Relative CMJ PF ($\text{N}\cdot\text{kg}^{-1}$)	0.291	0.289	0.779*	1.000*

PV = peak velocity; PP = peak power; PF = peak force.

*Correlations significant at $p \leq 0.05$.

reported a similar range of statistical power when assessing the correlations between similar variables. In our laboratory, test-retest reliability of maximal force assessed by an ISO squat and ISO mid-thigh is consistently $r \geq 0.98$, and test-retest reliability for CMJ peak power is consistently $r \geq 0.95$. Statistical analyses were performed using SPSS Version 12.0 (SPSS Inc., Chicago, IL).

RESULTS

The means and *SDs* for strength and power performance variables of the ISO squat, squat 1RM, ISO mid-thigh, power clean 1RM, and CMJ can be found in Table 1. When considering ISO and dynamic measures, which did not account for body mass (i.e., absolute strength) (Figure 1) and their relationships to both CMJ peak force and CMJ peak power, significant correlations existed. ISO squat peak force, squat 1RM, and power clean 1RM were all significantly correlated ($p \leq 0.05$) with CMJ peak force ($r = 0.639, 0.791$, and 0.840 , respectively). However, ISO mid-thigh peak force was not significantly correlated with CMJ peak force ($r = 0.538$). ISO squat peak force, ISO mid-thigh peak force, squat 1RM, and power clean 1RM were all significantly correlated with CMJ peak power ($r = 0.706, 0.750, 0.836, 0.856$, respectively). Further analysis revealed that there were no significant correlations between any of the 4 measures of absolute strength when compared to CMJ peak velocity and CMJ height. Results demonstrated that ISO squat peak force, ISO mid-thigh peak force, squat 1RM, and power clean 1RM were not significantly correlated with CMJ peak velocity ($r = -0.040, 0.079, -0.042$, and -0.216 , respectively). When considering CMJ height, ISO squat peak force, ISO mid-thigh peak force, squat 1RM, and power clean 1RM were not significantly correlated ($r = -0.073, 0.276, -0.219$, and 0.059 , respectively).

Further examination of data included analysis of the correlations between ISO and dynamic measures of relative strength to relative CMJ peak force, relative CMJ peak power, CMJ peak velocity, and CMJ height (Figure 2). Relative squat 1RM was found to be significantly correlated with relative CMJ peak power, CMJ peak velocity, and CMJ height ($r = 0.676, 0.731$, and 0.690 , respectively). Also, relative power

clean 1RM was significantly correlated with the same 3 CMJ variables ($r = 0.706, 0.698$, and 0.642 , respectively). However, no significant correlations were present when comparing relative squat 1RM to relative CMJ peak force ($r = 0.508$) or when comparing relative power clean 1RM to relative CMJ peak force ($r = 0.535$). When considering the relationships between relative ISO squat peak force and relative CMJ peak force, relative CMJ peak power, CMJ peak velocity, and CMJ height, no significant correlations were found ($r = -0.068, 0.269, 0.548$, and 0.276 , respectively). Further, relative ISO mid-thigh peak force was not significantly correlated with relative CMJ peak force, relative CMJ peak power, and CMJ peak velocity ($r = 0.095, 0.512$, and 0.544 , respectively). However, a significant correlation was present when comparing relative ISO mid-thigh peak force to CMJ jump height ($r = 0.588$).

Additional data analysis determined the relationships between RFD during ISO squat and ISO mid-thigh to CMJ performance variables (Figure 3). ISO squat RFD was found to significantly correlate with CMJ peak force and absolute CMJ peak power ($r = 0.721$ and 0.766 , respectively). However, ISO squat RFD was not significantly correlated with CMJ peak velocity or CMJ jump height ($r = -0.173$ and -0.045 , respectively). ISO mid-thigh RFD was found to significantly correlate with CMJ peak power ($r = 0.653$). However, ISO mid-thigh RFD did not significantly correlate with CMJ peak force, CMJ peak velocity, or CMJ jump height ($r = 0.458, 0.228$, and 0.352 , respectively).

Further, correlations between CMJ performance variables can be found in Tables 2 and 3. Most notable are the relationships of CMJ performance variables to CMJ height. CMJ peak velocity and relative CMJ peak power were both significantly correlated with CMJ height ($r = 0.826$ and 0.726 , respectively). No significant correlations were found between CMJ height and CMJ peak force, relative CMJ peak force, and CMJ peak power ($r = -0.116, 0.291$, and 0.266 , respectively).

DISCUSSION

The most significant finding from this investigation was that relative measures of multijoint dynamic strength best

correlate with measures of performance in a CMJ given the knee angles used. Absolute dynamic and ISO measures, which do not take into consideration body mass, do not significantly correlate with performance variables in the CMJ such as peak velocity and jump height; however, they do correlate with CMJ peak force and CMJ peak power. It appears that body mass must be accounted for by using a ratio of strength relative to body mass, and this is more clearly understood when considering Newton's second law (force = mass $\times$ acceleration) and the equation for power (power = force $\times$ velocity). Additionally, it seems that when comparing ISO and dynamic tests to CMJ performance, underlying neural and mechanical mechanisms may cause the dynamic tests to be more strongly correlated.

A comparison of absolute strength measurements to CMJ peak velocity and CMJ height did not reveal significant correlations. The findings in the current study are in agreement with previous investigations that did not find significant correlations between absolute measures of strength when compared to CMJ height (1,11,12,20,28,30). When discussing the ISO squat, our findings agree with previous investigations that found that peak force during the test does not significantly correlate with CMJ height (28,30). Additionally, when discussing the ISO mid-thigh, other investigations reported that peak force in the test was not significantly correlated with CMJ height (11,12). Furthermore, when considering absolute squat 1RM, our findings agree with those of Young and Bilby (31), who did not find a significant correlation with CMJ height. When comparing measures of relative strength to CMJ performance, it was discovered that relative squat 1RM and relative power clean 1RM were significantly correlated with relative CMJ peak power, CMJ peak velocity, and CMJ height. Previous investigations also demonstrated that relative squat 1RM is strongly correlated with CMJ height (7,30). To our knowledge, there is no evidence that suggests nonsignificant correlations between relative squat 1RM and CMJ height. Additionally, when body mass was taken into consideration, peak force in the ISO tests produced stronger correlations with CMJ peak velocity and CMJ height, and in the case of the ISO mid-thigh, a significant correlation with CMJ height was found. The latter finding has also been reported by Stone et al. (23).

In general, relative measures of strength, rather than absolute measures of strength, are more strongly correlated with CMJ height because relative measures account for the body mass, which is accelerated during a jump. This can be explained further by kinetic and kinematic equations that are commonly used in biomechanics. If Newton's second law is rearranged to solve for acceleration rate (acceleration = force $\div$ mass), where force is a subject's force output into a force plate and mass is a subject's body mass, the relationship between strength (i.e., the ability to produce force) and body mass becomes clearer. By increasing force output at a given body mass or by decreasing body mass and maintaining force output, the acceleration rate and velocity of a jump will be improved. The

importance of increased jump velocity to attain greater CMJ height has been established through previous investigations (9,10) and is supported by the significant correlation found between the 2 variables in this investigation. Furthermore, as velocity is increased during a jump, peak power output is also increased (8). Thus, resistance exercises such as the squat and power clean should be implemented to improve force, velocity, and subsequent power output during an explosive movement (i.e., jump, sprint). Due to the strong relationships between CMJ height and both relative dynamic strength and CMJ peak power, a concurrent strength and power training program, which includes the squat and power clean, appears to be appropriate for improving performance in lower body explosive exercises such as a vertical jump.

When compared to the dynamic measures of relative strength, the ISO measures did not correlate as strongly with CMJ performance. The validity of ISO tests to predict or correlate with performance in dynamic activity has been previously questioned because ISO tests are not specific to the dynamic movement patterns associated with human performance (3,25). Baker et al. (3) and a review by Wilson and Murphy (27) have both recognized the neural and mechanical mechanisms that differentiate ISO from dynamic tests. Previous investigations that studied the elbow flexors (19,23), elbow extensors, and knee extensors (2,17) have all reported that neural (i.e., motor unit) activation of the muscles is different in ISO and dynamic (concentric and/or eccentric) contractions. Babault et al. (2) and Kay et al. (17) have both suggested that differences in muscle length during the various types of contractions results in different neural activation. Kay et al. (17) went on to hypothesize that differences in muscle length in a concentric contraction may result in an afferent signal that is dissimilar to that which is stimulated in an ISO contraction. The difference in afferent signaling may potentially result in greater efferent command in a concentric contraction. It may be assumed that in the current study, neural activation during the muscle length changing eccentric and concentric phases of the CMJ was most closely associated with neural activation in the squat and power clean and as a result stronger correlations were found. Additional consideration has been given to the mechanical differences between ISO and dynamic activities. During dynamic activities, it is believed that the stretch-shortening cycle (SSC) allows for the utilization of stored elastic energy, and, as a result, concentric phase activity is enhanced (15). In this manner, squat 1RM and power clean 1RM are similar to the CMJ, while ISO tests are not similar. In addition to data that demonstrated nonsignificant correlations between ISO tests and CMJ height (1,11,12,20,28,30), further study has revealed that strength training, which only uses ISO contractions and thus no SSC, does not improve CMJ performance (18). However, ISO training does improve static jump performance, which also does not involve the SSC (18). This evidence further substantiates the idea and helps clarify evidence from the current investigation

that ISO tests may not be well correlated with dynamic activity because of the absence of the SSC.

In conclusion, this investigation demonstrated that relative multijoint dynamic measures of strength (squat 1RM and power clean 1RM) correlate best with CMJ performance variables (relative CMJ peak power, CMJ peak velocity, and CMJ height). Additionally, multijoint ISO tests were not well correlated with CMJ performance, but correlations did become stronger when body mass was taken into consideration. Thus, it appears that training programs should aim to improve both maximal strength and maximal power, while sustaining an optimal body mass in order to increase power production and improve vertical jump performance.

PRACTICAL APPLICATIONS

In this study, a significant correlation existed between CMJ peak power and CMJ height. This suggests that by increasing peak power output through training, CMJ height may be improved. Thus, training to maximize power becomes of interest to strength and conditioning professionals as CMJ height is often used as a tool to evaluate or project the talent of athletes at events such as the National Football League's Scouting Combine. In order to maximize power output from training, Cormie et al. (8) suggest that athletes train at the load that optimizes power output. For the jump squat and power clean, loads of 0 (i.e., body mass) and 80% of 1RM, respectively, have been found to optimize power output (8). Furthermore, in the current investigation, significant correlations were found between dynamic 1RM relative to body mass and CMJ peak power, CMJ peak velocity, and CMJ height. This suggests that increasing maximal strength while sustaining an optimal body mass may also help improve CMJ performance. However, in order to maximize gains in strength, the intensity of an exercise needs to be greater than that needed to optimize power (6,14). Thus, it seems that training programs that focus on maximizing both strength and power have the greatest potential for improving performance in explosive lower body movements. When compared to strength-only and power-only training programs, a concurrent strength and power training program has been found to be more effective in improving lower body strength and performance in lower body explosive movements such as vertical jumping and sprinting (14). Thus, it is recommended that strength and conditioning professionals implement concurrent strength-power training programs in order to maximize performance.

REFERENCES

- Anderson, MA, Gieck, JH, Perrin, D, Weltman, A, Rutt, R, and Denegar, C. The relationship among isometric, isotonic, and isokinetic concentric and eccentric quadriceps and hamstring force and three components of athletic performance. *J Orthop Sports Phys Ther* 14: 114-120, 1991.
- Babault, N, Pousson, M, Michaut, A, and Van Hoecke, J. Effects of quadriceps femoris muscle length on neural activation during isometric and concentric contractions. *J Appl Physiol* 94: 983-990, 2003.
- Baker, D, Wilson, G, and Carlyon, B. Generality versus specificity: a comparison of dynamic and isometric measures of strength and speed-strength. *Eur J Appl Physiol* 68: 350-355, 1994.
- Bemben, MG, Clasey, JL, and Massey, BH. The effect of the rate of muscle contraction on the force-time curve parameters of male and female subjects. *Res Q Exerc Sport* 16: 96-99, 1990.
- Blackburn, JR and Morrissey, MC. The relationship between open and closed kinetic chain strength of the lower limb and jumping performance. *J Orthop Sports Phys Ther* 27: 430-435, 1998.
- Campos, GER, Leucke, TJ, Wendeln, HK, Toma, K, Hagerman, FC, Murray, TF, Ragg, KE, Ratamess, NA, Kraemer, WJ, and Staron, RS. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. *Eur J Appl Physiol* 88: 50-60, 2002.
- Carlock, JM, Smith, SL, Hartman, MJ, Morris, RT, Ciroslan, DA, Pierce, KC, Newton, RU, Harman, EA, Sands, WA, and Stone, MH. The relationship between vertical jump power estimates and weightlifting ability: a field-test approach. *J Strength Cond Res* 18: 534-539, 2004.
- Cormie, P, McCauley, GO, Triplett, NT, and McBride, JM. Optimal loading for maximal power output during lower-body resistance exercises. *Med Sci Sport Exerc* 39: 340-349, 2007.
- Feltner, ME, Bishop, EJ, and Perez, CM. Segmental and kinetic contributions in vertical jumps performed with and without arm swing. *Res Q Exerc Sport* 75: 216-230, 2004.
- Feltner, ME, Franchetti, DJ, and Crisp, RJ. Upper extremity augmentation of lower extremity kinetics during countermovement jump vertical jumps. *J Sports Sci* 17: 449-466, 1999.
- Haff, GG, Carlock, JM, Hartman, MJ, Kilgore, JL, Kawamori, N, Jackson, JR, Morris, RT, Sands, WA, and Stone, MH. Force-time curve characteristics of dynamic and isometric muscle actions of elite women Olympic weightlifters. *J Strength Cond Res* 19: 741-748, 2005.
- Haff, GG, Stone, M, O'Bryant, HS, Harman, E, Dinan, C, Johnson, R, and Han, K. Force-time dependent characteristics of dynamic and isometric muscle actions. *J Strength Cond Res* 11: 269-272, 1997.
- Häkkinen, K. Force production characteristics of leg extensor, trunk flexor and extensor muscles in male and female basketball players. *J Sports Med Phys Fitness* 31: 325-331, 1991.
- Harris, GR, Stone, MH, O'Bryant, HS, Proulx, CM, and Johnson, RL. Short-term performance effects of high power, high force, or combined weight-training methods. *J Strength Cond Res* 14: 14-20, 2000.
- Ishikawa, M, Finni, T, and Komi, PV. Behaviour of vastus lateralis muscle-tendon during high intensity SSC exercises in vivo. *Acta Physiol Scand* 178: 205-213, 2003.
- Kawamori, N, Rossi, SJ, Justice, BD, Haff, EE, Pistilli, EE, O'Bryant, HS, Stone, MH, and Haff, GG. Peak force and rate of force development during isometric and dynamic mid-thigh clean pulls performed at various intensities. *J Strength Cond Res* 20: 483-491, 2006.
- Kay, D, St Clair Gibson, A, Mitchell, MJ, Lambart, MI, and Noakes, TD. Different neuromuscular recruitment patterns during eccentric, concentric and isometric contractions. *J Electromyogr Kinesiol* 10: 425-431, 2000.
- Kubo, K, Yata, H, Kanehisa, H, and Fukunaga, T. Effects of isometric training on the tendon stiffness and jump performance. *Eur J Appl Physiol* 96: 305-314, 2006.
- Murphy AJ and Wilson, GJ. Poor correlations between isometric tests and dynamic performance: relationship to muscle activation. *Eur J Appl Physiol* 73: 353-357, 1996.
- Nakazawa, K, Kawakami, Y, Fukunaga, T, Yano, H, and Miyashita, M. Differences in activation patterns in elbow

- flexor muscles during isometric, concentric and eccentric contractions. *Eur J Appl Physiol* 66: 214–220, 1993.
21. Paasuke, M, Ereline, J, and Gapeyeva, H. Knee extension strength and vertical jumping performance in Nordic combined athletes. *J Sports Med Phys Fitness* 41: 354–361, 2001.
 22. Paulus, DC, Reiser, RF II, and Troxell, WO. Pneumatic strength assessment device: design and isometric measurement. *Biomed Sci Instrum* 40: 277–282, 2004.
 23. Stone, MH, Sands, WA, Carlock, J, Callan, S, Dickie, D, Daigle, K, Cotton, J, Smith, SL, and Hartman, M. The importance of isometric maximum strength and peak rate-of-force development in sprint cycling. *J Strength Cond Res* 18: 878–884, 2004.
 24. Tax, AAM, Denier Van Der Gon, JJ, and Erkelens, CJ. Differences in coordination of elbow flexor muscles in force tasks and in movement tasks. *Exp Brain Res* 81: 567–572, 1990.
 25. Ugarkovic, D, Matavulj, D, Kukolj, M, and Jaric, S. Standard anthropometric, body composition, and strength variables as predictors of jumping performance in elite junior athletes. *J Strength Cond Res* 16: 227–230, 2002.
 26. Wilson, GJ and Murphy, JA. The use of isometric tests of muscular function in athletic assessment. *Sports Med* 22: 19–37, 1996.
 27. Winchester, JB, Erickson, TM, Blaak, JB, and McBride, JM. Changes in bar-path kinematics and kinetics after power-clean training. *J Strength Cond Res* 19: 177–183, 2005.
 28. Wisloff, U, Castagna, C, Helgerud, J, Jones, R, and Hoff, J. Strong correlation of maximal squat strength with sprint performance and vertical jump height in elite soccer players. *Br J Sports Med* 38: 285–288, 2004.
 29. Wisloff, U, Helgerud, J, and Hoff, J. Strength and endurance of elite soccer players. *Med Sci Sports Exerc* 30: 462–467, 1998.
 30. Young, W, Wilson, G, and Byrne, C. Relationship between strength qualities and performance in standing and run-up vertical jumps. *J Sports Med Phys Fitness* 39: 285–293, 1999.
 31. Young, WB and Bilby, GE. The effect of voluntary effort to influence speed of contraction on strength, muscular power, and hypertrophy development. *J Strength Cond Res* 7: 172–178, 1993.